Worksheet 6-6 B Review of 6-3 to 6-6

AP Calculus Name_____

The functions h, k, m, s and t are continuous. Use the information given to answer the following.

$$\int_1^7 h(x)\,dx = 8 \qquad \int_4^7 h(x)\,dx = 14$$

1) $\int_1^4 h(x)dx$

2) $\int_1^7 [2h(x) - 3]dx$

3) $\int_4^1 -5h(x)dx$

$$\int_8^{-2} k(x)dx = 5 \qquad \int_6^8 k(x)dx = 11$$

4) $\int_{-2}^6 k(x)dx$

5) $\int_8^6 [k(x) + 3]dx$

6) $\int_{-2}^8 7k(x)dx$

$$\int_{-3}^5 4m(x)\,dx = 12 \qquad \int_9^5 m(x)\,dx = -6$$

7) $\int_{-3}^9 m(x)dx$

8) $\int_5^9 [2 - m(x)]dx$

9) $\int_5^{-3} 3m(x)dx$

$$\int_2^{10} s(x)dx = 5 \qquad \int_7^{10} -2s(x)dx = 12 \qquad \int_2^7 t(x)dx = -8 \qquad \int_{10}^7 t(x)dx = 2$$

10) $\int_7^2 4s(x)dx$

11) $\int_2^{10} [s(x) - t(x)]dx$

12) $\int_7^2 [2s(x) - 3t(x) + 4]dx$

13.)

t seconds	0	2	4	9	14	15	16
$v(t)$ m/s	3	-2	7	12	2	3	-5

A bug is walking along a straight path for $0 \leq t \leq 16$ seconds. The velocity of the bug can be modeled by the differentiable function $v(t)$, in meters per second. Selected values of $v(t)$ are given in the table above.

A) Approximate $\int_0^{16} v(t)\,dt$ using a midpoint Riemann sum with the three subintervals indicated by the data above. Using correct units, interpret your answer in context of this problem.

B) Approximate $\int_4^{16} |v(t)|\,dt$ using a right Riemann sum with four subintervals indicated by the data in the table. Using correct units, interpret your answer in context of this problem.

C) Using a trapezoidal approximation with three subintervals, approximate $\frac{1}{9}\int_0^{9} v(t)\,dt$.

D) Approximate $a(3)$. Show the work that leads to your answer and include units of measure.